

hw 1

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1. (a) the sum over all x and y should add up to 1.

$$\sum_{x \in \{0,1,2\}} \sum_{y \in \{0,1\}} c(x+y) = 1 \quad (1)$$

this implies $c = \left(\sum_{x \in \{0,1,2\}} \sum_{y \in \{0,1\}} (x+y) \right)^{-1} c = 1/9$

(b)

$$P(x) = \sum_y c(x+y) = c(x+1) \quad (2)$$

$$P(y) = \sum_x c(x+y) = c(y+3) \quad (3)$$

(c)

$$P(x|y=1) = \sum_x c_2(x+1) \quad (4)$$

where $c_2 = 1/6$

(d) no. the distribution for x changes based on whether y is 0 or 1.

2. (a)

$$\begin{aligned} P_{z_i}(1) &= p \\ P_{z_i}(0) &= 1 - p \end{aligned} \quad (5)$$

(b) the pmf just follows a binomial distribution with $p = p$

$$P(x) = \binom{n}{x} p^x (1-p)^{n-x} \quad (6)$$

$$P(x \geq k) = \sum_{k}^n \binom{n}{x} p^x (1-p)^{n-x} \quad (7)$$

(c)